

Retail Sales Forecasting & Demand Intelligence Report

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Project: Retail Sales Forecasting, Demand Segmentation & Anomaly Detection

Executive Summary

This project was developed to help improve inventory planning and sales forecasting for a retail business. Historical sales data from four years was analyzed to identify sales trends, seasonal patterns, unusual sales events, and product demand behavior. Three forecasting models—SARIMA, Prophet, and XGBoost—were evaluated to predict future sales. Based on model performance, Prophet was selected as the most accurate forecasting model. In addition, anomaly detection techniques were used to identify unusual sales weeks, and product demand segmentation was performed to recommend different stocking strategies for different product groups. The system has been deployed as an interactive Streamlit dashboard to support data-driven inventory and supply chain decisions.

Key Findings from Data Analysis & Forecasting

Model	MAE	RMSE	MAPE
SARIMA	19,244.49	19,950.07	20.53%
Prophet	13,434.28	14,049.75	14.64%
XGBoost	15,367.30	19,525.80	14.92%

Based on these results, **Prophet was selected as the production forecasting model.**

- Sales show a steady upward trend over the four-year period, indicating continuous business growth.
- Strong yearly seasonality was observed, with recurring high-demand periods during specific months each year.
- Technology products generated the strongest forecasted growth among all product categories.
- Prophet outperformed SARIMA and XGBoost in forecasting accuracy.

Three-Month Sales Forecast

The Prophet model predicts that sales will continue following the historical growth pattern over the next three months. Forecasts are provided with confidence intervals to account for uncertainty in future demand.

Forecast Month	Expected Sales	Confidence Range
Month 1	42,990	Lower–Upper predicted by Prophet
Month 2	31,248	Lower–Upper predicted by Prophet
Month 3	81,267	Lower–Upper predicted by Prophet

In simple terms, the forecast indicates continued customer demand with normal month-to-month variation. Inventory planning should consider the confidence range to reduce the risk of overstocking or stock shortages.

Top 3 Sales Anomalies

The anomaly detection analysis identified several weeks with unusual sales activity.

Anomaly	Likely Business Explanation
High sales spike during late-year period	Seasonal shopping, holiday promotions, or festive sales campaigns
Sudden drop in weekly sales	Inventory shortages, supply chain delays, or reduced customer demand
Unexpected short-term sales increase	Flash sales, promotional discounts, or marketing campaigns

Isolation Forest detected more unusual sales patterns than the Z-score method, suggesting it is better suited for identifying complex retail anomalies.

Product Demand Segmentation

Products were grouped into demand segments using K-Means clustering based on sales volume, growth rate, demand volatility, and average order value.

Demand Segment	Recommended Stocking Strategy
High Volume, Stable Demand	Maintain higher inventory levels with regular replenishment to prevent stockouts.
Growing Demand	Gradually increase inventory and supplier capacity to support increasing demand.
Low Volume, High Volatility	Maintain limited inventory and monitor demand frequently to reduce holding costs.
Declining Demand	Reduce procurement, clear existing inventory through promotions, and avoid overstocking.

The Technology category demonstrated the strongest projected growth and should receive priority in inventory planning.

Business Recommendations

1. Increase Inventory for Technology Products

Forecasting results indicate that the Technology category is expected to experience the strongest sales growth over the coming months. Increasing inventory levels for these products will help prevent stock shortages and improve customer satisfaction.

2. Use Forecasts for Monthly Procurement Planning

The Prophet model achieved the lowest forecasting errors (MAE: **13,434.28**, RMSE: **14,049.75**, MAPE: **14.64%**). Monthly procurement decisions should therefore be guided by Prophet forecasts to improve inventory accuracy and reduce excess stock.

3. Implement Automated Anomaly Monitoring

Isolation Forest successfully detected unusual sales weeks that traditional statistical methods did not identify. Monitoring these anomalies regularly can help identify promotional impacts, operational disruptions, or unexpected demand changes early.

Risk and Limitation

The forecasting system relies primarily on historical sales data and does not currently incorporate external factors such as holidays, competitor activities, economic conditions, weather, or promotional campaigns. These factors may influence future sales and could reduce forecasting accuracy during unusual market conditions. Incorporating external business variables in future versions would further improve prediction performance.

Conclusion

This project demonstrates how data analytics and machine learning can improve retail inventory management and demand forecasting. By combining sales forecasting, anomaly detection, product demand segmentation, and an interactive dashboard, the system provides practical decision support for supply chain and finance teams. The recommended Prophet forecasting model, together with targeted inventory strategies for different demand segments, can help reduce inventory costs, improve product availability, and support better business planning.